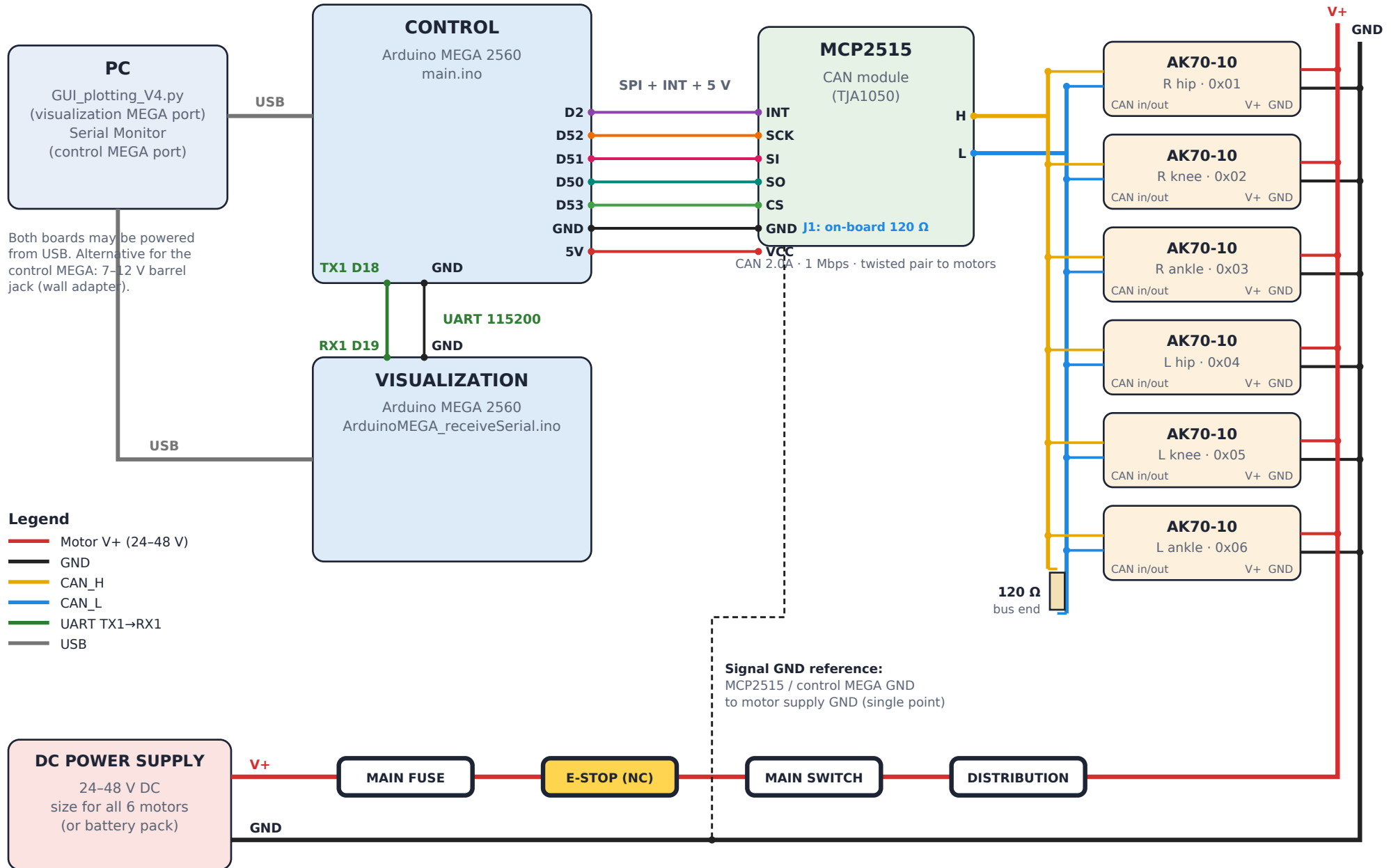


OpenRehEx — Wiring Diagram

Power, CAN bus, SPI and serial connections for 2 × Arduino MEGA 2560, MCP2515 and 6 × CubeMars AK70-10



Connection Tables & Checklist

Pin-to-pin connections, bus rules and power-safety notes

Control MEGA ↔ MCP2515 (SPI)

MCP2515	MEGA	Function
INT	D2	Interrupt (optional, CAN_INT_PIN = -1 if unused)
SCK	D52	SPI clock
SI	D51	MOSI
SO	D50	MISO
CS	D53	Chip select (CAN_CS_PIN)
GND	GND	Common ground
VCC	5V	Module supply (TJA1050 needs 5 V)

Control MEGA ↔ Visualization MEGA (UART)

Control	Visualization	Function
TX1 D18	RX1 D19	Telemetry 115200 baud
GND	GND	Mandatory common ground
RX1 D19	(not connected)	Link is one-way

USB

Board	Host	Use
Control MEGA	PC	Console + firmware upload
Visualization MEGA	PC	GUI data (set COM_PORT in the .py)

CAN bus

From	To	Note
MCP2515 H	CAN_H of motor 0x01 ... 0x06	daisy chain
MCP2515 L	CAN_L of motor 0x01 ... 0x06	daisy chain
120 Ω	MCP2515 end (J1 jumper)	termination 1
120 Ω	after the last motor (0x06)	termination 2
Check	≈ 60 Ω between H and L	power OFF

Use twisted pair for CAN_H/CAN_L, keep stubs short and run the bus as a line (no star). Each motor needs a unique ID set with CubeMarsTool before it joins the bus (all ship as 0x01). The original single-motor drawing (CAN_Protocol/CAN_connection.png) shows the same pinout.

Motor power

Net	Path
PSU V+	fuse → E-stop → switch → distribution → V+ ×6
PSU GND	distribution → GND ×6 (+ signal GND reference)
Wire gauge	size for the summed current of all motors
Connectors	use the motor's power connector; check polarity

Safety & power notes

- The E-stop must physically interrupt motor V+ and be reachable by the therapist. Firmware FAULT (damping) is not a safety function.
- Disabling or cutting power makes the joints back-drivable: the user must be supported (harness / parallel bars) before 'd' or E-stop.
- Back-driven motors regenerate energy. A lab PSU cannot absorb it and the bus voltage may rise (AK error 3 = over-voltage): prefer a battery, or add a braking resistor / TVS on the V+ bus, and keep the PSU voltage below the motor maximum.
- The 15 A supply in the README comes from the single-motor bench setup; re-size the PSU, fuse and wiring for six motors.
- Tie the MCP2515/Arduino GND to the motor supply GND at one point so CAN has a common reference; avoid ground loops through the PC.
- Verify pinouts against the datasheet/label of your AK70-10 revision before first power-up.